Register no:	2	0	2	3	1	1	8	0	5	2

Department of Information Science and Technology, Anna University, Chennai-25

Internal Assessment – I

V Semester – B. Tech (IT)

Regulation 2023

Course Code/Course Title: IT23503/ Compiler Design

Staff:Dr.P.Varalakshmi/Dr.Abirami Murugappan

Time: 90 minutes

Max Marks: 50

Date: 3-9-2025

CO1	Understand the concept of Lexical analysis to construct a Lexical Analyzer.
CO2	Understand the usage of Syntax Analysis to construct and use a parser appropriately.
CO3	Design and implement intermediate Code generator.
CO4	Understand the usage of Run time environment to develop applications.
CO5	Analyze and apply the code optimization and design a code generator.

BL – Bloom's Taxonomy Levels

(L1- Remembering, L2-Understanding, L3-Applying, L4- Analysing, L5-Evaluating, L6-Creating)

Q.No	PART- A (7 x 2 = 14 Marks)	Marks	CO	BL
1	State the alphabet and the language denoted by the following regular expressions. $a(a\|b)^*a$, $(0\|1)^+0(0\|1)(0\|1)?$	2	CO1	L2
2	Differentiate tokens, patterns and lexemes with suitable example	2	CO1	L1
3	Convert $(a \mid b)^*abb$ into Context Free Grammar	2	CO2	L2
4	Write a LEX program to identify the numbers and arithmetic operators (+,-,*,/)	2	CO1	L3
5	For the following grammar, give the rightmost derivation and parse tree for the string "aaabbabbba". $S \rightarrow aB / bA$ $A \rightarrow aS / bAA / a$ $B \rightarrow bS / aBB / b$	2	CO2	L3
6	Design a grammar for the set of strings of 0s and 1s with an equal number of 0s and 1s in the middle of the string.	2	CO2	L3
7	Apply left-factoring for the following grammar: S-> aSSbS/aSaSb/abb/b	2	CO2	L2
Q.No	**PART- B (2 x 12 = 24 Marks)**	**Marks**	**CO**	**BL**
8 (a)	Construct DFA for $(a\|b)^*a(a\|b\|c)$ by first converting regular expression to Epsilon-NFA, Epsilon-NFA to DFA. Further minimize the number of states in DFA. Check the validity of the strings like 'aac' and 'cc' in NFA and minimized DFA .	12	CO1	L3
	OR			
8(b)	Construct the minimum state DFA for $ab(a\|\varepsilon)^*c\|ac^*b$ by directly	12	CO1	L3

	converting regular expression to DFA using Syntax Tree approach.			
9(a)	Explain the phases of compiler in detail by simulating the results of the following expression in each phase of the compiler. result = a + b * c /d – 15 Where a, b, c, d are float data types.		CO2	L2
	OR			
9(b)	i. Explain various syntax error handling strategies ii. Draw suitable transition diagrams to recognize the given password which consists of digits, special characters and alphabets.	6 6	CO2 CO1	L2
Q.No	**PART- C (1 x 12 = 12 Marks)**	**Marks**	**CO**	**BL**
10.	Construct the predictive parser for the following grammar. You may eliminate the left recursion if required. bexp -> bexp or bterm | bterm bterm -> bterm and bfactor | bfactor bfactor -> not bfactor | (bexp) | true | false Parse the following sentences for the above grammar. **false or true and not false**	12	CO2	L3

2/9/2025

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